

By: AKH
 Rev.--- By: ---
 Chk: ---

Date Jun 2026
 Date ---
 Date ---

MacTest

Macaulay Beam Analysis - Example Worksheet

102 - (2) EQ Span - Point Load

Beam Sect = W14x61 W14x90
 A = 17.90 26.50 in²
 I = 640.00 999.00 in⁴

E = 29,000,000 lb / in²

Length Units in
 Force Units lb

Beam Segments (First Segment Begins at 0)

	Segment No.	X End	A	I	E
		in	in ²	in ⁴	lb / in ²
DO NOT DELETE ROWS (GRAPHED)	1	600.00	17.90	640.00	2.900E+07
	2				
	3				
	4				
	5				

Supports

	Supp. No.	X	Type	Imposed Movement	
		in	---	* dY (vert)	** dr (rot)
DO NOT DELETE ROWS (GRAPHED)	1	0.0	P	0.0	0.000
	2	300.0	P	0.0	0.000
	3	600.0	P	0.0	0.000
	4				
	5				

* (settlement) ** (Not Working)

Distributed Loads (downward loads entered as NEGATIVE) - UNFACTORED

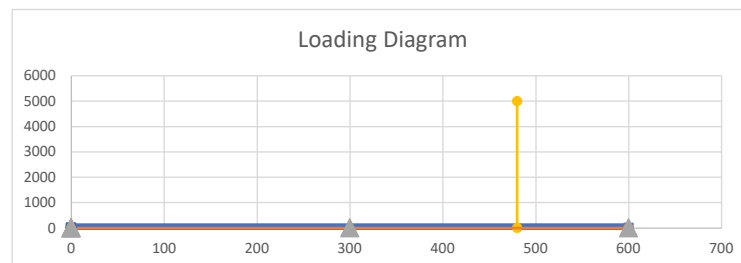
	Load No.	x1	x2	w1 (at x1)	w2 (at x2)
		in	in	lb / in	lb / in
DO NOT DELETE ROWS (GRAPHED)	1	0.0	600.0	0.0	0.0
	2	0.0	0.0	0.0	0.0
	3	0.0	0.0	0.0	0.0
	4	0.0	0.0	0.0	0.0
	5	0.0	0.0	0.0	0.0

Load	
lb / ft	
0.0	
0.0	
0.0	
0.0	

Point Loads - UNFACTORED

	Load No.	X	Py	My
		in	lb	lb-in
DO NOT DELETE ROWS (GRAPHED)	1	480	-5000.00	0.00
	2	0	0.00	0.00
	3			
	4			
	5			

My Load	
lb-ft	
0.0	
0.0	
0.0	
0.0	



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BEAM REACTIONS

Position (X)	Enforced Disp (dy)	Enforced Rot (theta)	Vert Reac (Ry)	Internal Moment (Mz)	Type
(L)	(L)	(rad)	(F)	(F-L)	(---)
0.00	0.0	0.0	-420.0	0.0	P
300.00	0.00	0.00	2,840.1	0.0	P
600.00	0.00	0.00	2,580.0	0.0	P

Vert Reac
(kip)
-0.420
2.840

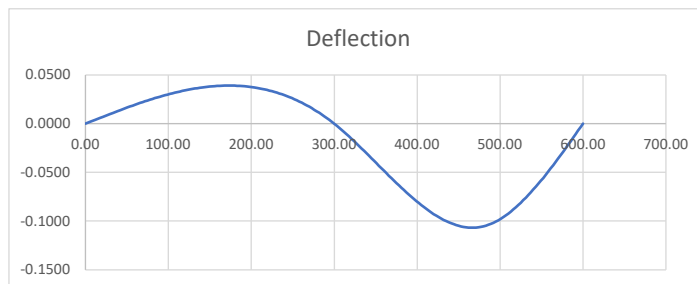
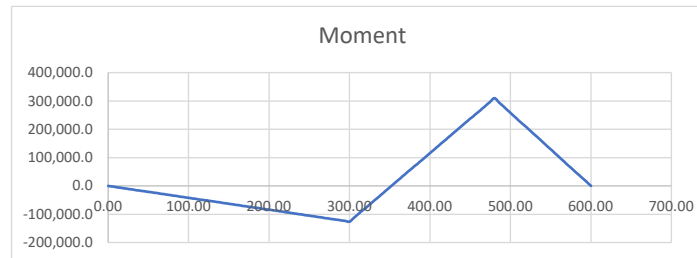
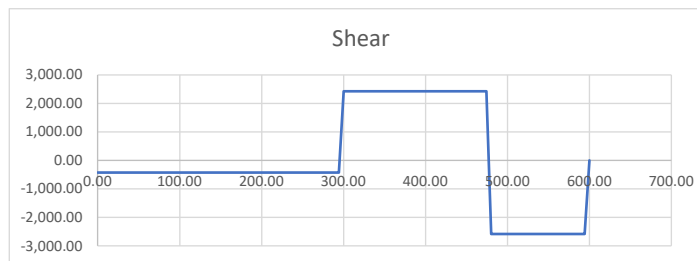
BEAM REACTIONS EQUILIBRIUM CHECK

Item	Units	Loads	Reactions	Residual	Check Sum
Sum Fy	F	-5,000.0	5,000.0	0.0	OK
Sum M @ x=0	F-L	-2.4000E+06	2.4000E+06	0.0	OK

BEAM MAX/MIN LOAD EFFECTS

Item	Position (X)	Shear (V)	Moment (M)	Defl. (delta)
(---)	(L)	(F)	(F-L)	(L)
Max +V	300.00	2,420.0	-126,012.6	0.0000
Max -V	480.00	-2,580.0	309,595.0	-0.1054
Max +M	480.00	-2,580.0	309,595.0	-0.1054
Max -M	300.00	2,420.0	-126,012.6	0.0000
Max Defl	468.00	2,420.0	280,554.5	-0.1068

Max/Min Value
(---)
2.42 kip
-2.58 kip
25.80 kip-ft
-10.50 kip-ft
-0.107 in



INPUT SUMMARY

General Info

Beam Length:	50 ft
Section Name:	W14x61
Self Weight:	False

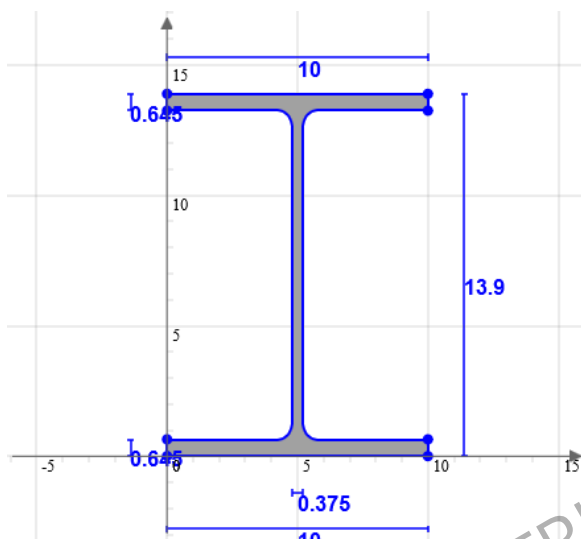
Supports

Support Type	Location
Pinned	0 ft
Roller	50 ft
Roller	25 ft

Loads

Load Type	Location	Magnitude	Load Case
Point Load	40 ft	-5000 lb	DL

Beam Section



Geometric Properties		
A	17.9	in ²
C _z	5	in
C _y	6.95	in

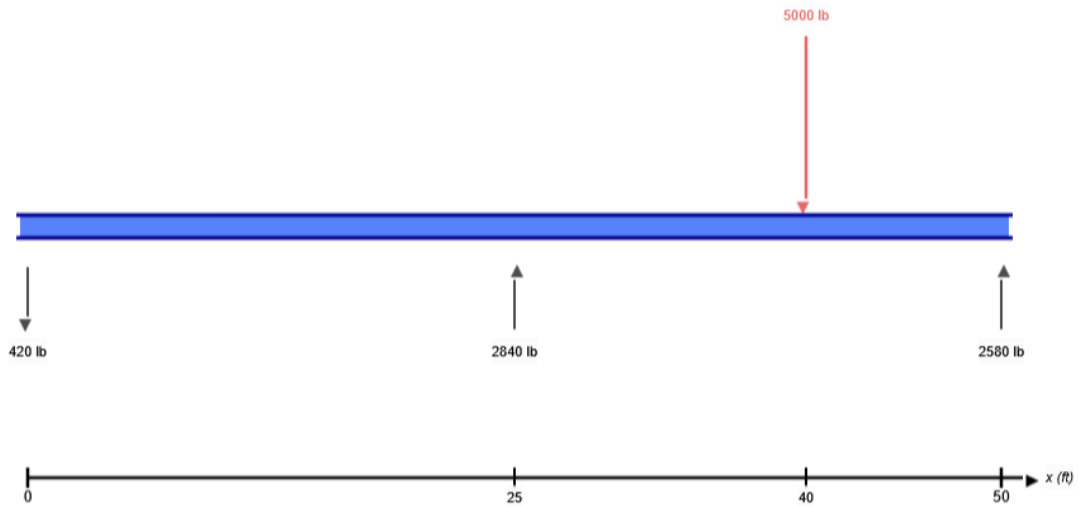
Bending Properties		
I _z	640	in ⁴
I _y	107	in ⁴

Shear Properties		
A _z	11.277	in ²
A _y	4.99	in ²

Torsion Properties		
J	2.19	in ⁴
r	0.991	in

Shape	Material	E (ksi)	v	ρ (lb/ft ³)
W14x61	Structural Steel	29000	0.27	490

FREE BODY DIAGRAM



RESULT SUMMARY

Check	Status	Limit	Ratio	Max
Deflection	PASS	L/250	0.044	L/5648
Custom Stress Limit	PASS	39 ksi	0.086	3.362 ksi
Material Yield	PASS	36 ksi	0.093	3.362 ksi
Material Strength	PASS	58 ksi	0.058	3.362 ksi

ANALYSIS RESULTS

Reactions

Support at	R_x (lb)	R_y (lb)	M (kip-ft)
0	0	-420	0
25	0	2840	0
50	0	2580	0

Force Extremes

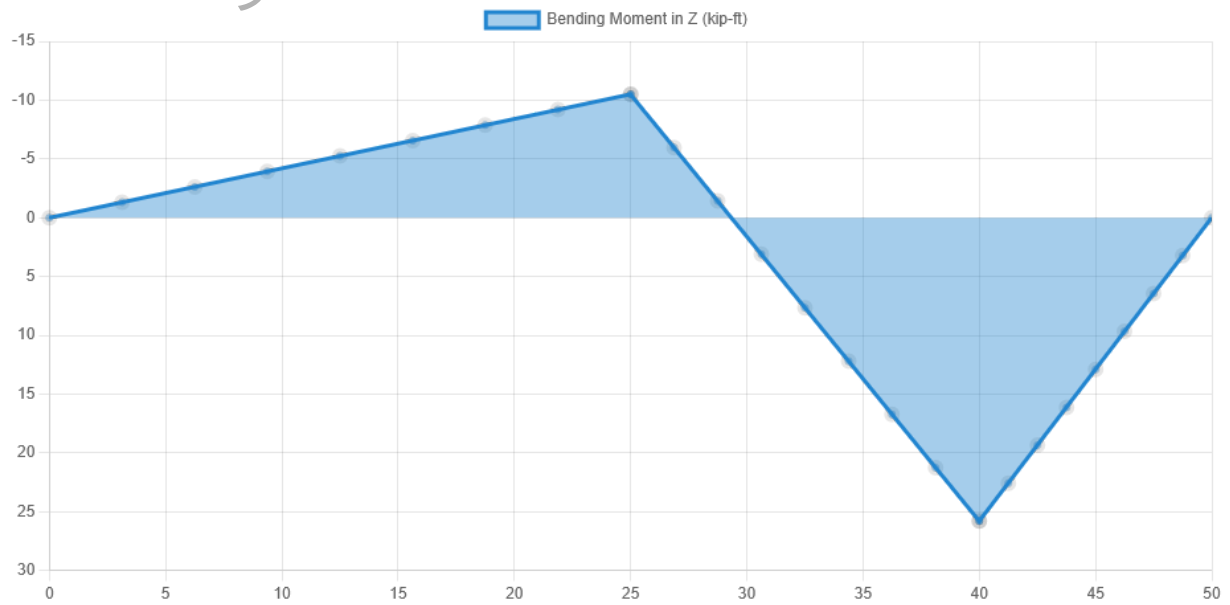
Result	Max	Min
Bending Moment	25.8 kip-ft	-10.5 kip-ft
Shear	2420 lb	-2580 lb
Displacement Y	0.039 in	-0.106 in

Stress Extremes

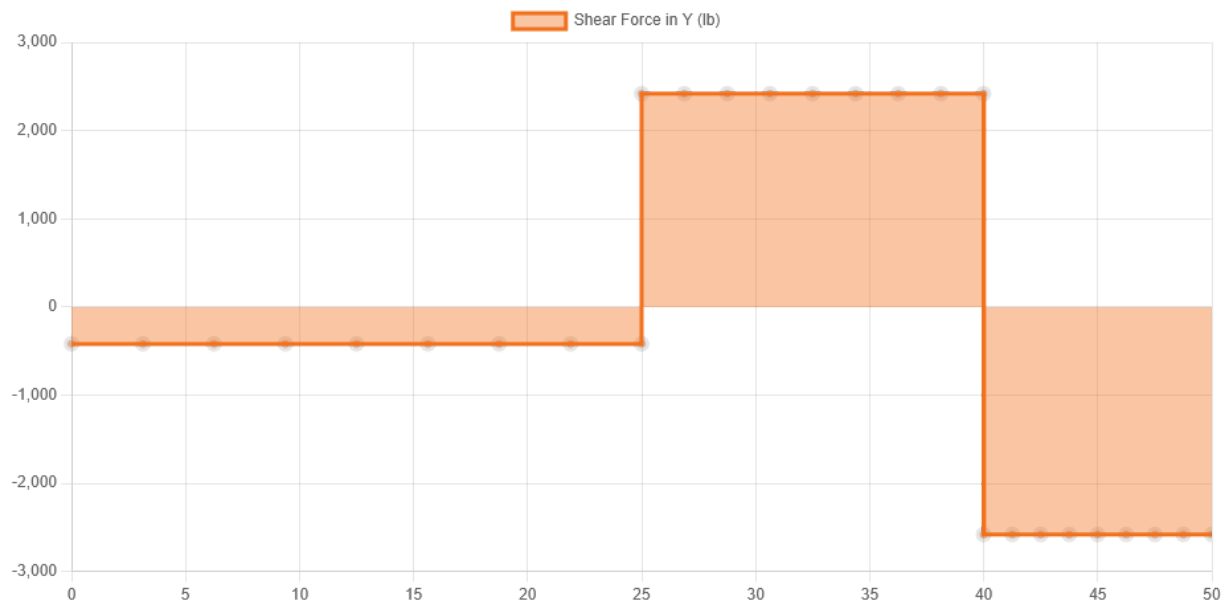
Result	Max (ksi)	Min (ksi)
Bending Stress	3.362	-3.362
Shear Stress Total	0.551	0.09
Max Combined Normal Stress	3.362	0
Min Combined Normal Stress	0	-3.362

DIAGRAMS

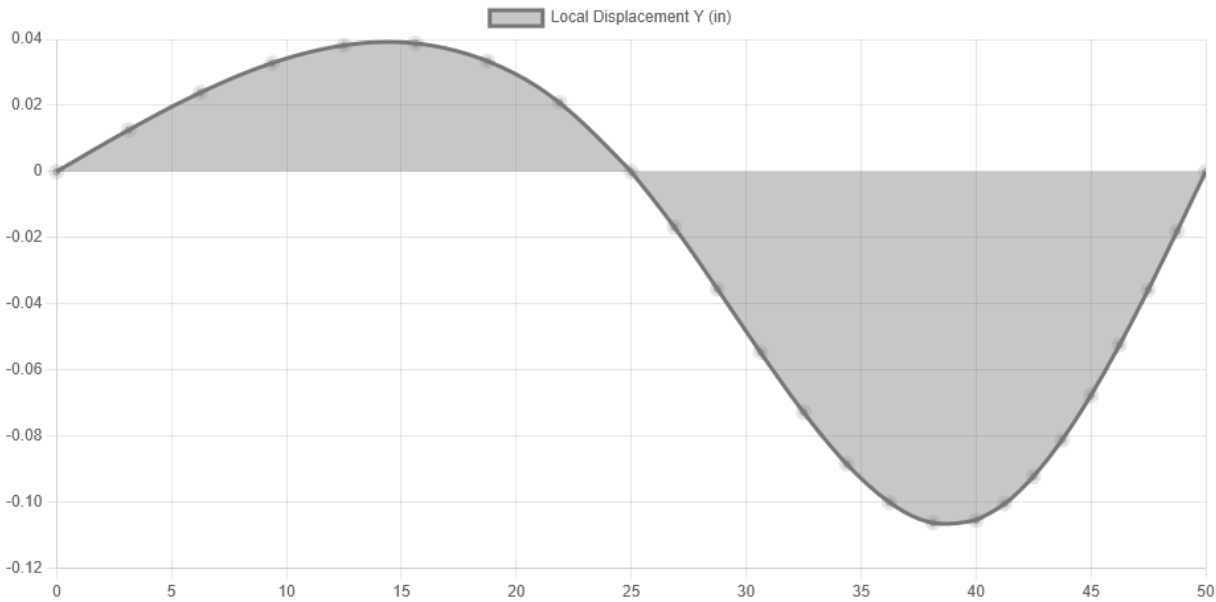
Bending Moment Diagram



Shear Force Diagram



Displacement



Location (ft)	0	15.625	25	38.125	50
Total Deflection (in)	0 ⓘ	0.039 ⓘ	0 ⓘ	0.106 ⓘ	0 ⓘ
Deflection/Span	-	L/7735	-	L/2824	-
ⓘ The Deflection/Span results are calculated using the analysis results and the Deflection Limit of L/250 set in the model settings.					